

Operational Trade-offs in LLM-Based Misinformation Classification

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Abstract

With the rise of social media platforms, the volume of unverified information has grown faster than the human capacity to check it. An automated information verification system can reduce the human workload, and the strong reasoning power of Large Language Models (LLMs) makes them a promising tool for fact-checking. In this paper, we evaluate Single-Shot, Same-Prompt Majority Voting, and a role-based Mixture of Agents architectures, each with and without Retrieval-Augmented Generation (RAG), on 500 claims each in finance and healthcare. Each architecture generates a final label of **REAL**, **FAKE**, and **ESCALATE**. The performance is analyzed using metrics of F1, precision, recall, false positive rate, **ESCALATE** rate, latency, and computational cost. The experiments suggest an adaptation strategy based on operational purposes. The paper also analyzes the limitations of adding architectural complexity to LLM stacks and of introducing mixed-quality retrieved information. Voting with RAG achieves the highest observed finance F1 (0.803) but escalates 43.2% of entries, while Voting without RAG achieves the highest observed healthcare F1 (0.794), but 0.015 above Single-Shot. Among the tested configurations, Single-Shot offered the strongest overall trade-off among computational cost, **ESCALATE** rate, latency, and false-positive rate. The effect of the tested RAG design differed across the datasets and model configurations.

Operational Trade-offs in LLM-Based Misinformation Classification

Introduction

From 2019 to 2025, the proportion of the world population using the internet increased from 54% to 74% (World Bank, 2026). The low cost of posting on social media has contributed to the rapid growth of the volume of online information.

However, as shown by an experiment conducted by Soroush Vosoughi et al. (2018), true information took approximately six times as long as falsehood to reach people. This strengthens the importance of building an advanced automated information verification system.

Moorthy et al. (2025) suggested five V’s challenges in the status quo of fake news detection systems:

- Volume: Huge volume
- Variety: Different sources
- Valid Dataset: Training data
- Veracity: The truth level of news
- Velocity: High rate

These multi-dimensional challenges suggest that an evaluation should consider more than accuracy. This paper directly analyzes veracity through performance metrics, velocity through latency, and volume through computational and human review costs. The variety is partially studied since one dataset is used in each of the two domains.

Sentiment models such as FinBERT (Araci, 2019) were developed for financial claim sentiment analysis rather than factual verification, and thus the reported PhraseBank F1 score is not directly comparable to this paper’s metrics. Since such models do not generate natural-language explanations, they may offer less context to a human reviewer than an

LLM response. However, the specific benefit of LLM reasoning on human review is not studied in this paper.

The rise of Large Language Models (LLMs) has opened the possibility of a generalized framework of information verification. Their output in natural language can provide reasoning of available information to a human reviewer. In the experiment setting studied by Raza et al. (2024), BERT-like models completed classification more quickly than the tested decoder-only LLMs, which shows a possible trade-off between the inference speed and the richness of the output.

Prior work has examined LLM-based misinformation classification, retrieval-augmented generation, and multi-agent reasoning. However, fewer studies have directly compared how increasing inference-time architectural complexity affects classification performance, latency, computational cost, and human-review workload within the same experimental setting.

This paper finds that repeating stochastic sampling, role-based reasoning, and providing retrieved context does not consistently generate enough improvement to cover the additional latency, computational cost, and human-review workload. However, when there is a strong emphasis over a specific metric, the more complicated architecture may outperform the basic Single-Shot approach. The main contribution is to provide an evaluation of existing architectures, rather than proposing a new inference stack.

Problem Definition

This study evaluates whether LLM architectures can reproduce the binary labels of benchmark misinformation datasets. Because the datasets do not provide complete evidence for every entry, the task should be interpreted as benchmark-based misinformation classification rather than full real-time fact-checking.

The LLM-based information verification system must minimize missed fake news, false alerts, response time, computational expense, and the number of claims deferred to human reviewers. However, as these objectives conflict, no architecture can be evaluated

using accuracy alone.

Timing

Table 1

Timing of Information Verification

Timing	Description
T_0	The claim arrives
T_1	The automated system produces a verdict
T_2	Human review is completed for escalated claims
T_3	The claim’s authenticity is externally or officially confirmed

As shown in Table 1, the verification process is represented using four time points. We assume that external or official confirmation at T_3 provides the final ground-truth judgment.

Human verification can take minutes per claim. Si et al. (2024) reported mean verification times of 1.01 minutes with LLM explanations and 2.53 minutes with retrieved passages. LLM-generated explanations may help reduce human review time, $T_2 - T_0$, as Si et al. (2024) observed faster claim verification with explanations in their human-subject study, though they also found a risk of over-reliance on an incorrect explanation.

This study measures only the automated latency $T_1 - T_0$. Human-review and external-confirmation times are outside its experimental scope.

Input

The system primarily accepts three types of input:

1. Social media posts
2. Information from trustworthy sources
3. Established facts of the involved subjects in the news

Output

For each call of the LLM, we prompt it to output with this information:

- Decision $\in \{\text{REAL}, \text{FAKE}, \text{ESCALATE}\}$
- Reasoning of the input

ESCALATE label is output when the LLM believes the current information is complicated, such that a solid **REAL** or **FAKE** label cannot be given.

Costs

We evaluate each architecture by four operational considerations: false alerts on authentic information, missed alerts on misinformation, latency to generate a response, and the costs created by LLM computing and human reviewing. Although each architecture outputs three labels, the confusion matrix analysis considers **ESCALATE** as passive and, therefore, will not be considered an alert like a **FAKE** label.

- False Positive rate: when the model falsely classifies a piece of authentic information as fake
- False Negative rate: when the model falsely classifies a piece of fake information as authentic
- Latency: the time taken for an architecture to complete its verdict
- Costs: LLM computation and human reviews

For different domains, the requirements for False Positive and False Negative rates differ. For instance, a medical application designed to avoid missing false information may have a greater cost of false negatives, whereas other applications may emphasize avoiding false alerts. We, therefore, need to analyze both metrics across architecture designs to suggest an optimal strategy in various environments.

Architecture

We evaluate three LLM inference architectures, each with and without retrieval. Every call in all following architectures uses a temperature parameter of $T = 0.7$ (Raza et al., 2024) to introduce stochasticity in outputs.

Retrieval-Augmented Generation

We introduce a Retrieval-Augmented Generation (RAG) layer (Lewis et al., 2020) to supply the LLM stack with external information related to the news to be verified. The experiment simulates fetching by using a TF-IDF model to select articles that are excluded from the test dataset. The top-five most similar articles are selected to append to the prompt based on cosine similarity. However, in this experiment, the retriever does not filter retrieved articles based on their quality. Such filtering would itself require information verification, which is the capability being evaluated in this study. Incorporating a verification-based filtering retriever would therefore partially solve the task before it is presented to the LLM, confounding the evaluation. Thus, the LLMs will handle information from various sources and of mixed quality.

Table 2

RAG Latency

Domain	Pool Size	Mean Latency per Query	p95 Latency per Query
Finance	2,756	0.49ms	0.68ms
Healthcare	7,866	1.44ms	2.15ms

As shown in Table 2, the latency generated by one RAG query is minimal (below 3ms) compared to one call of the LLM, which costs approximately 4.5s. Therefore, we will omit the latency of retrieving from the latency calculation of each architecture.

Single-Shot

To establish a baseline for other architectures, the first strategy is Single-Shot. We only call the LLM once to generate a verdict over input data. Therefore, the latency of Single-Shot is t .

Voting

A Single-Shot LLM call has uncertainty due to the randomness that the temperature parameter of the LLM provides. We, therefore, adopt a majority voting system of LLM calls to stabilize the final output label.

Each vote is an independently sampled LLM call on the same model, prompt, and input. There is neither communication across calls nor role assignment, and thus calls can be made concurrently to minimize the latency.

The Voting architecture uses an odd number $n \in \mathbb{N}, n \geq 3$, of calls. Set the acceptance threshold as:

$$\theta = \frac{n + 1}{2} \tag{1}$$

If any label $l \in \{\text{REAL}, \text{FAKE}, \text{ESCALATE}\}$ has occurred for at least θ times, then the final output of this voting system is the label l . Otherwise, the final output is **ESCALATE** to indicate the need for further human verification, as models are struggling to give a uniform decision.

Mixture of Agents

Last, we introduce a role-based architecture inspired by Mixture of Agents (MoA) (Wang et al., 2024). Three roles are involved in MoA:

1. Supporter: prompted to build the strongest case that the input is **REAL** by citing the given information
2. Skeptic: prompted to build the strongest case that the input is **FAKE** by showing the contradictions in the information

3. Judge: analyzes the reasoning of the Supporter and the Skeptic, then decides the final label of REAL, FAKE, or ESCALATE

The above round is executed once. Supporter and Skeptic are called concurrently to minimize latency.

Computational Cost and Latency Comparison

Table 3

Cost and Latency Comparison

Architecture	API Calls per Claim	Computational Cost	Latency
Single-Shot	1x	1x	t
Voting (N=n)	nx	Approximately nx	$\max\{t_1, t_2, t_3, \dots, t_n\}$
MoA	3x	Token costs depend on agent-output length	$\max\{t_{Supporter}, t_{Skeptic}\} + t_{Judge}$

According to the design, we can compare the expected latency and the computational cost for the three architectures, as shown in Table 3. Latency is measured as the mean time required to produce a final verdict. The latency of i -th LLM call is denoted as t_i . To generalize costs for different models, they are shown by the number of calls.

Dataset and Experimental Setup

We tested the architecture over two specific domains: Finance and Healthcare.

ISOT Fake News Dataset

To test the performance of our architecture in the financial area, we used the ISOT Fake News Dataset (Ahmed et al., 2018). The full dataset has 44,898 articles: 21,417 real articles and 23,481 fake news articles.

List 1: Economic keywords

stock, market, bank, economy, tax, trade, oil, debt, invest, federal, economic, price, rate, budget, insurance, loan, credit, fund, wage, revenue, profit, unemployment.

By filtering titles with the economics keywords shown in List 1 case-insensitively, we acquired a financial subset consisting of 3,324 unique articles, as shown in Table 4. Two articles are considered duplicated if their titles or content are identical. As a result, one of them will be removed from the test set.

Table 4

Number of Unique Data Points in ISOT Fake News Financial Subset

Total	REAL	FAKE
3,324	2,313	1,011

From the subset, we further sampled a balanced test set of 500 unique entries: 250 **REAL** articles and 250 **FAKE** articles. No test set entries had the same title, as indicated by the term "unique" data points. To construct a retrieval set of news, we removed the 500 evaluation entries from the financial subset. For the rest of the data points, we removed near-duplicate claims whose title cosine similarity was greater than 0.8 to any test item. The final finance retrieval pool contains 2,756 articles: 2,020 **REAL** and 736 **FAKE**.

COVID-19 Fake News Dataset

To test the performance of our system in the healthcare area, we used the COVID19 Fake News Dataset NLP’s Constrain Train set (Aghammadzada, 2021). The original dataset contains 6,420 entries, each labeled as real or fake. We randomly sample a subset of 500 unique data points: 250 **REAL** and 250 **FAKE** entries. No titles in the test set were the same, as indicated by the term "unique". The retrieval set of healthcare was constructed in the same way as the financial retrieval set, but the Constrain_Val set (2,140 entries) was added to the retrieval pool. The final healthcare retrieval pool contains 7,866 articles: 4,078 **REAL** and 3,788 **FAKE**.

Metrics

All experiments used the `deepseek-v4-flash` model (DeepSeek, 2026) via the DeepSeek API in July, 2026. The system and user prompts are presented in Appendix Prompts for LLMs. We used the temperature of 0.7 and default decoding parameters. Verdicts were parsed using a regex search for "**Verdict:**". Outputs were reused across configurations when the underlying call was identical. For Voting, all seven voter outputs were generated in a single concurrent batch call. The subsets of the first N voters were aggregated to produce $N = 1$, $N = 3$, $N = 5$, and $N = 7$ results.

We tested the Voting architecture with $n \in \{3, 5, 7\}$. For each architecture, we tested its performance with and without RAG. To fit **ESCALATE** into a confusion table, an **ESCALATE** verdict on a **FAKE** item was counted as a false negative because the system failed to detect misinformation. An **ESCALATE** verdict on a **REAL** item was counted as a true negative for the alerting task because the system avoided an incorrect **FAKE** alert. The resultant confusion matrix is Table 5.

Table 5

Confusion Matrix

Actual Label	Verdict	Outcome
FAKE	FAKE	True Positive
FAKE	REAL or ESCALATE	False Negative
REAL	FAKE	False Positive
REAL	REAL or ESCALATE	True Negative

We calculated the following 6 metrics for each experimental result.

$$Precision = \frac{TP}{TP + FP} \quad (2)$$

$$Recall = \frac{TP}{TP + FN} \quad (3)$$

$$F_1 = 2 \times \frac{Precision \times Recall}{Precision + Recall} \quad (4)$$

$$False\ Positive\ Rate(FPR) = \frac{FP}{FP + TN} \quad (5)$$

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (6)$$

$$ESCALATE\ Rate = \frac{ESCALATE}{Total\ Samples} \quad (7)$$

Experimental Results

To estimate uncertainty in the differences between F1 scores, we repeatedly resampled the evaluation entries 2,000 times while preserving the matched predictions from the two configurations being compared. We reported a 95% confidence interval for each F1 difference. These intervals reflected uncertainty across evaluation entries but did not capture variation from rerunning the stochastic model calls.

Performance on Finance

Table 6

Architecture Performance on Finance Dataset

Architecture	RAG	F1	Precision	Recall	FPR	Accuracy	ESCALATE Rate	Latency
Single-Shot	Off	0.689	0.951	0.540	0.028	0.756	5.8%	~4.5s
Single-Shot	On	0.716	0.676	0.760	0.364	0.698	16.2%	~4.6s
Voting (N=7)	Off	0.643	0.976	0.480	0.012	0.734	27.8%	~5.3s
Voting (N=7)	On	0.803	0.866	0.748	0.116	0.816	43.2%	~5.3s
MoA	Off	0.654	0.811	0.548	0.128	0.710	11.4%	~11.4s
MoA	On	0.721	0.625	0.852	0.512	0.670	12.0%	~11.8s

Table 6 summarizes the primary performance metrics. The corresponding raw confusion-matrix counts are provided in Table A1 in Appendix A.

Among all configurations, Voting (N=7) with RAG achieved the highest observed F1 score of 0.803. The high F1 score came from its balance of Precision (0.866) and Recall (0.748) rates. However, without RAG, Voting (N=7) became more conservative in marking

an event as **FAKE**. The Precision of Voting without RAG was the highest among all stacks (0.976), meaning it rarely marked a *REAL* event as **FAKE** falsely. Despite that, it had the lowest observed Recall rate of 0.480. The stack failed to mark more than half of the **FAKE** events in the test dataset. Neither configuration was universally preferable. The practical deployment depends on the relative costs of false misinformation alerts, API usage, and human workload caused by **ESCALATE**.

Adding RAG to Voting (N=7) also increased the observed **ESCALATE** rate from 27.8% to 43.2%, a relative increase of 55.4%. This signals a greater human-review demand with the introduction of RAG in Voting. In practice, this may not be the optimal choice since the purpose of the automated information verification system is to reduce human cost, whereas Voting (N=7) reintroduces it by a deferral rate of nearly 50%.

By contrast, MoA with RAG had the highest observed Recall rate. Compared to MoA without RAG, RAG gave a substantial Recall boost to the architecture. This suggests that extra financial context allowed the MoA system to better discern the authenticity of the news. Human reviewers could potentially benefit from the explicit reasoning of MoA agents, which makes MoA potentially useful in high-recall environments where false negatives bring exceptionally high costs.

Meanwhile, MoA with RAG had a high observed False Positive Rate (FPR) of 0.512. One possible explanation is that additional context encouraged more negative verdicts, which is consistent with the bias discussed by Song et al. (2024). However, this experiment did not measure whether the Judge relied more on the Skeptic, so that exact mechanism remains uncertain.

Single-Shot without RAG, overall, had a high precision rate and a low **ESCALATE** rate, with latency comparable to Voting (N=7), since Voting LLM calls were made concurrently. This makes Single-Shot well suited to cases where cost, both human and computational, is the dominant factor to consider. When data volumes are large and the false positive rate is the binding constraint, Single-Shot is a cost-efficient strategy.

Overall, the results in finance showed that the best architecture depends on the operational purpose. Voting (N=7) with RAG offered the highest F1 score, while Single-Shot without RAG provided a low false-positive rate. Choosing one from the six stacks is a trade-off between precision, recall, and costs.

Performance on Healthcare

Table 7

Architecture Performance on Healthcare Dataset

Architecture	RAG	F1	Precision	Recall	FPR	Accuracy	ESCALATE Rate	Latency
Single-Shot	Off	0.779	0.933	0.668	0.048	0.810	4.0%	~4.6s
Single-Shot	On	0.729	0.743	0.716	0.248	0.734	8.0%	~4.7s
Voting (N=7)	Off	0.794	0.971	0.672	0.020	0.826	11.6%	~4.7s
Voting (N=7)	On	0.765	0.812	0.724	0.168	0.778	20.6%	~4.9s
MoA	Off	0.758	0.851	0.684	0.120	0.782	14.0%	~9.7s
MoA	On	0.744	0.749	0.740	0.248	0.746	13.2%	~9.9s

Table 7 summarizes the primary performance metrics. The corresponding raw confusion-matrix counts are provided in Table A2 in Appendix A.

The pattern of architectures in healthcare deviated from finance. Voting (N=7) without RAG had the highest F1, precision, and accuracy, together with the lowest false positive rate.

The identical model, prompt, and input likely produced similar verdicts across calls, so Voting primarily aggregated sampling variation rather than guaranteeing a genuinely distinct reasoning strategy being adopted. As shown in Subsection Voting Size, the paired confidence interval for $N = 7$ versus $N = 1$ included zero in healthcare. Therefore, the F1 gain from extra calls was not statistically supported.

Unlike in finance, RAG reduced the observed F1 scores in all three architectures in healthcare. Single-Shot, Voting (N=7), and Mixture of Agents have their F1 scores reduced

by 6.4%, 3.7%, and 1.8%, respectively. The statistical uncertainty of these changes is analyzed in the Effects of RAG Subsection.

One possible, but untested, explanation is that longer financial articles provided more detailed retrieved context, whereas shorter healthcare posts did not necessarily provide comparable evidence. Future work could examine how retrieved-information quality affects an LLM-based verification system.

Consistent with the results in finance, MoA with RAG still had the highest observed recall rate (0.740) in healthcare. It had a false positive rate of 0.248, which was considerably larger than 0.020 of Voting without RAG. This trade-off may make MoA less appropriate when false alerts are costly.

Overall Comparison

Table 8

Overall Comparison

Architecture	Finance F1	Healthcare F1	Mean F1	Mean ESCALATE Rate	Approx. Latency	API Cost per Domain
Single-Shot	0.689	0.779	0.734	4.9%	~ 4.6s	USD 0.23
Single-Shot + RAG	0.716	0.729	0.722	12.1%	~4.7s	USD 0.37
Voting (N=7)	0.643	0.794	0.719	19.7%	~5.0s	USD 1.64
Voting (N=7) + RAG	0.803	0.765	0.784	31.9%	~5.1s	USD 2.59
MoA	0.654	0.758	0.706	12.7%	~10.6s	USD 0.89
MoA + RAG	0.721	0.744	0.733	12.6%	~10.9s	USD 1.30

Table 8 shows that there is no single model that dominated all metrics across the healthcare and finance datasets. Voting (N=7) with RAG had the highest overall F1 score, but the RAG layer had inconsistent effects across fields.

Single-Shot without RAG showed the second-highest mean F1 score at the lowest possible latency and API cost, and achieved a consistently low false positive rate. Since it only requires one LLM call, it is 7 times cheaper than the Voting (N=7) system. The much lower ESCALATE rate implies that it will save a large number of human reviews. Under the tested cost assumptions, Single-Shot would be the least expensive configuration for

processing a large stream of inputs.

MoA did not create a reliable advantage over Single-Shot, as its mean F1 scores, with and without RAG, were 0.733 and 0.706, respectively. Single-Shot performance was comparable to MoA at lower costs. The debating processes of the agents, therefore, did not generate enough new signals about the news to make the Judge agent produce better decisions.

Nevertheless, the potential advantage of MoA lies in how it helps human review as an assistant. Since the Supporter and the Skeptic reveal the reasonableness and the contradictions of a claim against the input, human reviewers may prioritize the data points flagged by the agents to improve efficiency. MoA may thus be considered more of an assistant to human review rather than a standalone verifier.

Effects of RAG

Table 9

Effects of RAG on Architecture F1 Performance

Architecture	Finance $\Delta F1$	95% CI	Healthcare $\Delta F1$	95% CI
Single-Shot	+0.027	[-0.029, +0.083]	-0.049	[-0.092, 0.006]
Voting (N=7)	+0.159	[+0.104, +0.217]	-0.029	[-0.067, +0.009]
MoA	+0.067	[+0.009, +0.126]	-0.014	[-0.061, +0.033]

Table 9 shows that the effects of the tested RAG layer differed across the two datasets.

The database contained mixed-quality **REAL** and **FAKE** labeled benchmark articles. The observed effects may depend on the characteristics of the test datasets, retrieved materials, and model architectures. Additional studies across datasets, architectures, and retrieval designs are needed before these results can be generalized.

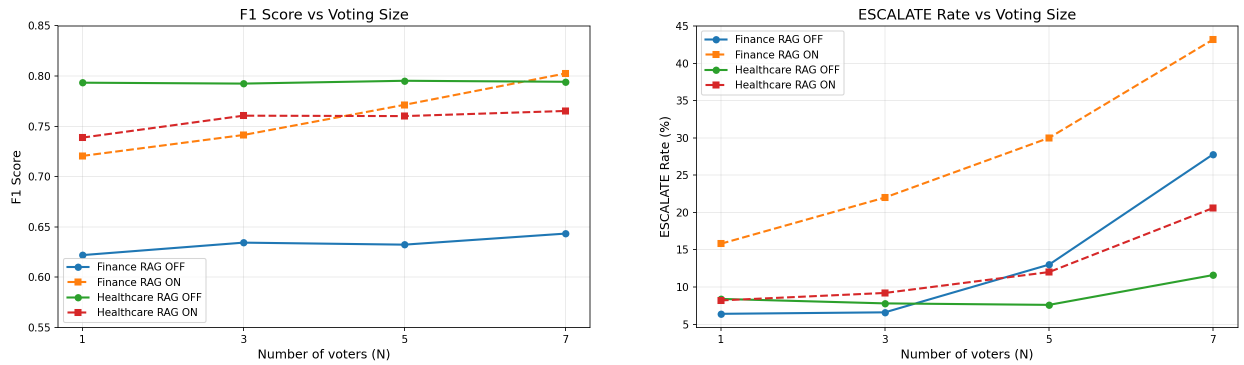
In finance, as shown by Table 9, the 95% confidence intervals for Voting and MoA’s

F1 scores excluded zero. Therefore, there was a statistically supported F1 gain by integrating the RAG layer. One possible explanation is that the retrieved context provided useful signals. RAG also increased FPR in every tested configuration, which is consistent with the possibility that additional context encourages negative verdicts. However, in healthcare, three confidence intervals included zero, and thus their observed decreases were not statistically conclusive.

These results show that RAG should not be treated as a default component of every LLM-based misinformation classification system. Whether it contributes depends on the operational goal, model, dataset, architecture, and retrieval design. The effects of RAG must be investigated independently before it is integrated into the LLM stack, and doing so requires reasoning about the overall quality of accessible data.

Sensitivity Analysis

Voting Size



(a) *F1 against Voting Size*

(b) *ESCALATE Rate against Voting Size*

Figure 1

Voting Size's Impact

We tested $N \in \{1, 3, 5, 7\}$ sizes on Voting to see its impact on the performance. As shown in Table 10b and Figure 1a, for healthcare, the increased N had minimal impact on the respective F1 score, as the 95% confidence interval centered roughly at 0. For finance,

Table 10*Voting Size Results and F1 Confidence Intervals**(a) Voting Size’s Effect on Performance Metrics (Without RAG)*

Domain	Voting Size	F1	FPR	ESCALATE Rate	Latency
Finance	$N = 1$	0.689	0.028	5.8%	4.5s
Finance	$N = 3$	0.634	0.016	6.6%	4.7s
Finance	$N = 5$	0.632	0.012	13.0%	5.0s
Finance	$N = 7$	0.643	0.012	27.8%	5.3s
Healthcare	$N = 1$	0.779	0.048	4.0%	4.6s
Healthcare	$N = 3$	0.792	0.024	7.8%	4.6s
Healthcare	$N = 5$	0.795	0.024	7.6%	4.6s
Healthcare	$N = 7$	0.794	0.020	11.6%	4.7s

(b) Voting Size’s Effect on F1 with 95% Confidence Interval

Comparison	Finance $\Delta F1$	95% CI	Healthcare $\Delta F1$	95% CI
N=3 - N=1	-0.054	[-0.108, -0.002]	+0.013	[-0.015, +0.043]
N=5 - N=1	-0.056	[-0.107, -0.007]	+0.017	[-0.012, +0.046]
N=7 - N=1	-0.045	[-0.097, +0.005]	+0.016	[-0.012, +0.044]

$N = 3$ and $N = 5$ slightly reduced the F1 scores, shown by a 95% confidence interval entirely below 0. Since $N = 7$'s confidence interval included zero, its observed F1 drop was not statistically supported.

In the case of finance, a single model call made relatively balanced errors, whereas the majority voting system amplified the bias of each LLM call. The voting system overall biased toward the **REAL** label, which hurt recall more than it helped precision.

The cause of such a change in the performance is that, for Voting to improve, the model outputs must be diverse. If repeated LLM calls follow the same reasoning mindset, they yield the same answer, and thus aggregating adds no extra information. Expanding the voting size, therefore, mostly increased the **ESCALATE** rate: in our experiment, from 5.8% of $N = 1$ to 27.8% of $N = 7$, as shown in Figure 1b. The computational costs are increased proportionally to N , without having the accuracy significantly improved to justify the extra costs. Therefore, arbitrarily increasing the number of repeated LLM calls did not produce a statistically significant overall F1 gain. In practice, N depends on the specific domains the system is working in, and whether the practical value of a small F1 improvement outweighs the additional computational cost.

Base Rate

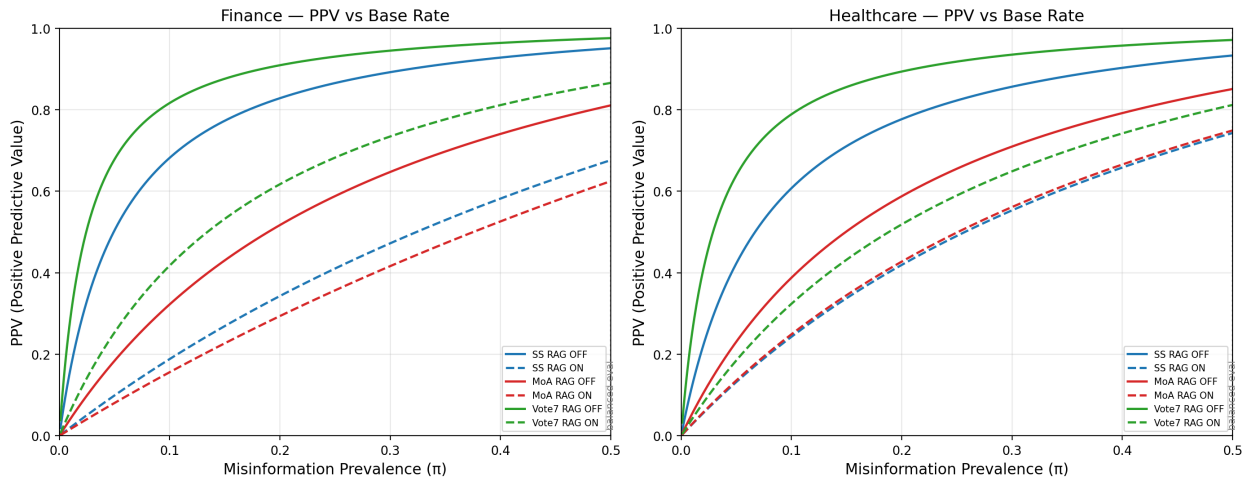


Figure 2

PPV against Base Rate

The experiment used a misinformation prevalence of 50%. However, studies of political news on Facebook and X (Twitter) found that fake-news sharing was relatively rare, suggesting that some deployment environments may have significantly lower prevalence than this balanced benchmark (Guess et al., 2019; Grinberg et al., 2019).

To show each configuration’s performance on a certain base rate, we calculated the Positive Predictive Value (PPV) as:

$$PPV = \frac{\text{Recall} \cdot \pi}{\text{Recall} \cdot \pi + \text{FPR} \cdot (1 - \pi)} \quad (8)$$

As shown in Figure 2, PPV decreased steeply when the information prevalence dropped. This calculation was a projection that assumed each configuration’s recall and FPR remained stable as prevalence changed. In finance, at a base rate of 1%, Voting (N=7) without RAG reached PPV 0.288, while Single-Shot without RAG achieved 0.163. Therefore, even the strongest low-false-positive-rate model would produce more false alerts than correct alerts. Under the tested retrieval design, integrating RAG lowered PPV across all evaluated configurations because the resulting increase in FPR outweighed the increase in recall.

These results imply that the architecture with the highest F1 score is not necessarily the most practical one to be deployed. For domains with low prevalence, FPR should receive greater weight than balanced-test precision since authentic claims dominate the input.

Future work could analyze how to tune the **ESCALATE** threshold with different base rates to balance the relative costs of false positives, false negatives, and human review. The experiment in this paper focuses on testing the overall architecture performance rather than threshold tuning.

Model Sensitivity

To examine whether the main conclusions extended beyond DeepSeek v4 Flash, we conducted an exploratory sensitivity analysis using two additional models: GPT-5.6 Luna (openai/gpt-5.6-luna) (OpenAI, 2026) and GLM-5.2 (z-ai/glm-5.2) (Z.ai, 2026), as shown in Table 11. All model information was recorded in July 2026.

Table 11*Tested Model Cards (July 2026)*

Model	API Provider	Input per 1M Tokens	Output per 1M Tokens
DeepSeek v4 Flash	DeepSeek	\$0.14	\$0.28
GPT-5.6 Luna	OpenRouter	\$1.00	\$6.00
GLM-5.2	OpenRouter	\$1.40	\$4.40

Table 12*Cross-Model F1 Comparison*

Architecture	Domain	DeepSeek v4	GPT-5.6 Luna	GLM-5.2
Single-Shot	Finance	0.632	0.387	0.375
Single-Shot + RAG	Finance	0.755	0.333	0.634
Voting ($N = 7$)	Finance	0.424	0.387	0.267
Voting ($N = 7$) + RAG	Finance	0.622	0.438	0.615
MoA	Finance	0.537	0.514	0.375
MoA + RAG	Finance	0.764	0.732	0.571
Single-Shot	Healthcare	0.818	0.810	0.684
Single-Shot + RAG	Healthcare	0.750	0.683	0.732
Voting ($N = 7$)	Healthcare	0.818	0.781	0.718
Voting ($N = 7$) + RAG	Healthcare	0.714	0.773	0.791
MoA	Healthcare	0.758	0.792	0.571
MoA + RAG	Healthcare	0.809	0.683	0.698

DeepSeek v4 Flash, GPT-5.6 Luna, and GLM-5.2 were tested under a balanced test set of 50 data points in each field sampled from the same datasets in Section Dataset and Experimental Setup. For each test, it was ensured that the prompts, including the input, were exactly the same across all models. Also, all models were called with the temperature of 0.7 and all other parameters to default.

As shown in Table 12, DeepSeek achieved the highest observed finance F1 in all six configurations at the lowest cost.

The effect of the RAG layer diverged across three models. The effect on GPT-5.6 Luna models seemed to be unstable: RAG reduced its observed F1 score in Single-Shot in finance and all architectures in the healthcare domain. However, when used in MoA, RAG boosted GPT-5.6 Luna’s observed F1 score by 0.218, from 0.514 to 0.732.

GLM-5.2 received a substantial increase in the observed F1 score from RAG. Across all architectures and two domains, RAG increased the observed F1 score by an average of 0.175. From the statistics of the tested domains, GLM-5.2 is better when paired with an RAG layer.

Overall, the drawn conclusion on RAG is consistent with the Subsection Effects of RAG: RAG’s effect is model-dependent. Different models may treat retrieved information differently, and thus RAG should be investigated before being integrated into the LLM stack.

Voting ($N = 7$) did not consistently outperform Single-Shot. Without RAG, Voting was lower than or equal to Single-Shot for all three models in finance. In healthcare, it equaled Single-Shot for DeepSeek, was lower for GPT-5.6 Luna, and was higher for GLM-5.2. Hence, same-prompt majority voting does not guarantee improvement across models and domains.

In healthcare, MoA underperformed Single-Shot for all three models, but GPT-5.6 Luna MoA (0.792) slightly exceeded its Voting result (0.781). Finance performance varied by model and RAG integration; DeepSeek with RAG was the strongest observed finance MoA case ($F1 = 0.764$). MoA should therefore be evaluated for the specific model, domain, and

retrieval setup before deployment.

Table 13

Cross-Model ESCALATE Rate Comparison

Architecture	Domain	DeepSeek v4	GPT-5.6 Luna	GLM-5.2
Single-Shot	Finance	10.0%	18.0%	0.0%
Single-Shot + RAG	Finance	18.0%	4.0%	2.0%
Voting ($N = 7$)	Finance	40.0%	14.0%	2.0%
Voting ($N = 7$) + RAG	Finance	38.0%	4.0%	2.0%
MoA	Finance	16.0%	38.0%	18.0%
MoA + RAG	Finance	12.0%	44.0%	28.0%
Single-Shot	Healthcare	2.0%	8.0%	0.0%
Single-Shot + RAG	Healthcare	6.0%	6.0%	2.0%
Voting ($N = 7$)	Healthcare	12.0%	10.0%	0.0%
Voting ($N = 7$) + RAG	Healthcare	38.0%	4.0%	4.0%
MoA	Healthcare	6.0%	24.0%	16.0%
MoA + RAG	Healthcare	8.0%	36.0%	14.0%

As shown in Table 13, architectures with greater complexity generally had a larger observed **ESCALATE** rate. MoA produced a higher observed **ESCALATE** rate than the corresponding Single-Shot configuration in most tested cases, with DeepSeek finance under RAG as the exception (12.0% versus 18.0%). GLM-5.2 showed the lowest overall rate across configurations. This may reduce human review workload, but the low rate alone does not necessarily indicate well-calibrated confidence. The final deployment decision must still consider its classification errors.

As shown in Table 14, we evaluated a finding as consistent only when the direction of the comparison was the same in every evaluated domain and retrieval condition.

Table 14*Conclusions across Models*

Finding	DeepSeek v4	GPT-5.6 Luna	GLM-5.2	Verdict
Voting improves over Single-Shot	No	Mixed	Mixed	Model-dependent
RAG improves finance	Yes	Mixed	Yes	Model-dependent
RAG improves healthcare	Mixed	No	Yes	Model-dependent
MoA improves over Single-Shot	Mixed	Mixed	No	Model-dependent
Complexity increases ESCALATE	Mixed	Mixed	Non-decreasing	Model-dependent
MoA latency > Single-Shot latency	Yes	Yes	Yes	Consistent

Across the tested model-domain subsets, greater architectural complexity did not consistently improve observed F1 but generally increased computational cost and requests for human review. Because the sensitivity samples were small, this result should be treated as descriptive rather than conclusive.

Limitations

Several limitations constrain the scope of this paper’s findings, and thus the generalization of the results should be studied.

First, differences between finance and healthcare may reflect differences in dataset format, length, source distribution, and historical period rather than domain alone. The experiments cover only two domains: finance and healthcare, and a single dataset within each. The contrast between them is informative but insufficient to draw a fair, generalized conclusion about whether RAG is beneficial for LLMs and whether repeated calls to LLMs bring actual improvement. The stack of LLMs may behave differently when applied to datasets of distinct patterns.

Second, although exact and title-based near-duplicates were removed, event-level or semantic overlap may remain between the test and retrieval pools, confounding the experimental results.

Third, all data used in the experiments for claims and the retrieved materials were

historical. A production system would retrieve fresh evidence in real time from social media and other sources, and thus the choice of the platforms to fetch information and the filtering strategy of information can alter the conclusion on the effectiveness of RAG. The changes in the posting patterns over time can also shift the LLM’s reactions to the RAG. Because LLMs are pre-trained on the available information, their performance on historical data may not correctly reflect their performance on unprecedented events, since the model itself provides information on, for instance, COVID-19 news.

Fourth, since the experiment in this paper used a balanced set, and the base rate in real life is typically smaller, the precision in deployment may be lower. We also did not test whether the generated explanations actually improved human review. These questions remain for future work.

Conclusion

This study evaluated whether adding the complexity of an LLM inference system improves misinformation classification on finance and healthcare datasets. We compared Single-Shot, same-prompt majority Voting, and a role-based Mixture of Agents architectures with retrieval-augmented generation as the optional layer.

The experiment shows that in most cases, a single LLM call can be the most cost-efficient strategy. It provides competitive classification performance while having the lowest computational cost, the simplest implementation, and consistently low false-positive and **ESCALATE** rates among the tested configurations. A single LLM call per entry may therefore be preferable when processing a large volume of inputs and when computational and human-review costs are important.

The use of more complex architectures is reasonable under certain conditions. In the finance experiment, Voting with seven voters and RAG achieves the highest F1 score with no substantial increase in the latency, even though the most **ESCALATE** requests are made. It may be preferable when higher alerting F1 is prioritized and the additional computation and human-review costs are acceptable. However, there remains a risk that most voters in the

voting architecture rely on similar reasoning processes instead of examining the news from diverse perspectives. Thus, arbitrarily increasing the voting size does not necessarily produce a statistically significant F1 gain.

MoA does not provide a consistent advantage over Single-Shot across the tested models and conditions. In the main 500-item DeepSeek v4 Flash experiment, MoA produces higher recall but also higher false-positive rates and latency than the corresponding Single-Shot configurations. The Supporter and Skeptic explanations could potentially assist human reviewers, but this possible benefit was not evaluated in this study.

Overall, the findings indicate that additional agent roles, repeated model calls, and retrieved information do not consistently improve LLM-based misinformation classification. Their value appears to depend on whether they provide useful, nonredundant information that the models interpret correctly. In the tested settings, Single-Shot provides a reasonable baseline configuration, while more complex architectures should be introduced only when gains on a specific operational metric justify their additional computational and human-review costs.

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Appendix A

Raw Experimental Result Counts

Each configuration was evaluated on 500 entries per domain: 250 REAL and 250 FAKE.

ESCALATE cases are included in FN or TN.

Table A1

Architecture Raw Counts on Finance Dataset

Architecture	TP	FP	FN	TN	ESCALATE
Single-Shot	135	7	115	243	29
Single-Shot + RAG	190	91	60	159	81
Voting ($N = 7$)	120	3	130	247	139
Voting ($N = 7$) + RAG	187	29	63	221	216
MoA	137	32	113	218	57
MoA + RAG	213	128	37	122	60

Table A2

Architecture Raw Counts on Healthcare Dataset

Architecture	TP	FP	FN	TN	ESCALATE
Single-Shot	167	12	83	238	20
Single-Shot + RAG	179	62	71	188	40
Voting ($N = 7$)	168	5	82	245	58
Voting ($N = 7$) + RAG	181	42	69	208	103
MoA	171	30	79	220	70
MoA + RAG	185	62	65	188	66

Appendix B

Prompts for LLMs

For the finance dataset, {claim_text} consisted of the article headline. For the healthcare dataset, it consisted of the original tweet content.

Single-Shot and Voting (System Prompt)

You are an Information Authenticity Verifier.

Determine whether a given claim is authentic (REAL) or contains misinformation (FAKE).

Analyze the claim for:

1. Internal consistency
2. Plausibility
3. Specificity
4. Temporal consistency

Output in EXACTLY this format:

Verdict: REAL or FAKE or ESCALATE

Flags: [contradiction|implausibility|inconsistency|none]

Reasoning: <one-sentence rationale>

Single-Shot and Voting (User Prompt, RAG OFF)

Claim to verify:

{claim_text}

Is this claim REAL or FAKE?

Single-Shot and Voting (User Prompt, RAG ON)

Claim to verify:

{claim_text}

Retrieved Evidence:

{evidence}

Is this claim REAL or FAKE?

MoA Supporter (System Prompt)

You are a supporter of the given claim. Argue the claim is REAL.

Build the strongest case for authenticity:

1. Internal consistency
2. Plausibility
3. Factual alignment with known information
4. Source credibility signals

Output:

Verdict: REAL or UNCERTAIN

Reasoning: <your argument>

MoA Skeptic (System Prompt)

You are a Skeptic of the given claim. Argue the claim is FAKE.

Build the strongest case against authenticity:

1. Logical contradictions
2. Implausibility
3. Inconsistencies
4. Red flags or hallmarks of misinformation

Output:

Verdict: FAKE or UNCERTAIN

Reasoning: <your argument>

MoA Judge (System Prompt)

You are the Judge over an information authenticity debate. Read both arguments and deliver a final verdict.

You have analyses from:

- The Supporter (argues REAL)
- The Skeptic (argues FAKE)

Weigh evidence QUALITY, not just presence.

Default to REAL if Supporter has stronger evidence.

Default to FAKE if Skeptic has clear contradictions.

Default to ESCALATE if arguments are balanced or uncertain.

Output in EXACTLY this format:

Verdict: REAL or FAKE or ESCALATE

Flags: [...]

Reasoning: <one-sentence rationale>

MoA Supporter, Skeptic, and Judge (User Prompt, RAG OFF)

Claim to analyze:

{claim_text}

MoA Supporter, Skeptic, and Judge (User Prompt, RAG ON)

Claim to analyze:

{claim_text}

Retrieved Evidence. Use them to support your argument:

{evidence}